

Noise Trader Risk in Financial Markets

J. Bradford De Long · Andrei Shleifer · Lawrence H. Summers · Robert J. Waldmann

AUTHORS	J. Bradford De Long, Andrei Shleifer, Lawrence H. Summers, Robert J. Waldmann
PUBLICATION YEAR	1990
JOURNAL	The Journal of Political Economy, vol. 98, no. 4, pp. 703–738
DOI	10.1086/261703
URL	https://www.journals.uchicago.edu/doi/abs/10.1086/261703
RESEARCH AREA	Behavioral Finance · Asset Pricing under Limited Arbitrage
SERIES	Behavioral Finance
PAPER TYPE	Theoretical, with applied interpretation of documented empirical anomalies
LITERATURE REVIEW	Arthur Faugeron
DATE REVIEWED	23 August 2026

ABSTRACT

The paper builds a two-period overlapping-generations model in which a fixed mass of irrational noise traders, whose beliefs about next-period prices are unpredictable, trade alongside risk-averse rational arbitrageurs against a riskless substitute asset. Because arbitrageurs have short horizons, they cannot wait out a reversal, so the unpredictability of noise traders' future opinions becomes a priced risk in its own right, distinct from any risk in the asset's dividends. This "noise trader risk" limits arbitrage, allows prices to diverge persistently from fundamental value even when nothing fundamental is uncertain, and can let noise traders earn higher expected returns than the sophisticated investors who trade against them, because noise traders bear a disproportionate share of a risk that their own unpredictability creates. The authors derive a closed-form pricing function, decompose the expected-return gap between the two investor types into four offsetting effects, extend the model to a dynamic setting in which new investors imitate whichever strategy recently performed best, and argue that the mechanism accounts for excess volatility, short-run mean reversion in returns, the closed-end fund discount, and the Mehra-Prescott equity premium.

1 Introduction

Milton Friedman's defense of speculation rests on a single mechanical claim: an investor who trades on mistaken beliefs must, on average, buy high and sell low relative to the investors who trade on correct beliefs, so mistaken traders lose money and are eventually priced out of the market by the very arbitrageurs who exploit them (Friedman 1953). Fama (1965) supplies the empirical companion to this claim in the efficient-markets literature: because rational arbitrageurs stand ready to absorb any mispricing created by uninformed trading, prices should track fundamental value closely and irrational demand should leave no lasting trace. De Long, Shleifer, Summers, and Waldmann (hereafter DSSW) do not dispute that arbitrageurs are rational. They dispute the implicit assumption that does the real work in Friedman's argument: that betting against irrational traders is riskless, or at least that the only risk involved is risk about the asset's fundamentals, which a diversified, patient arbitrageur can bear costlessly.

The paper's device for exposing this assumption is deliberately spare. It removes fundamental risk from the model entirely — both of the paper's two assets pay an identical, certain dividend — so that if Friedman's argument is correct, arbitrage should drive the risky asset's price to exact equality with the safe asset's price, leaving no role whatsoever for irrational "noise traders." It does not. Because arbitrageurs are risk averse and short-lived, and because noise traders' beliefs evolve unpredictably from one period to the next, an arbitrageur who buys an underpriced asset today must accept the risk that noise traders become even more pessimistic before he is forced to sell. That risk is priced, it deters aggressive arbitrage, and it survives even though every agent in the model, sophisticated and irrational alike, holds a portfolio with literally zero exposure to any real economic risk. The paper's central theoretical claim is that this "noise trader risk" is sufficient, on its own, to explain why prices can diverge persistently from fundamentals and why noise traders need not be selected out of the market by their losses.

The question the paper poses matters for a specific reason: the entire theoretical case for informational efficiency in asset markets, as developed through the 1960s and 1970s, depends on arbitrage being unconstrained enough to absorb any demand that is not information-based. If arbitrage is instead constrained by a risk that has nothing to do with fundamentals and everything to do with the unpredictability of other traders' irrationality, then prices can be systematically wrong for reasons that are neither a market failure in the usual sense nor a violation of individual rationality among the arbitrageurs themselves. Prior to this paper, two distinct literatures had chipped at Friedman's argument without dislodging its core claim. Figlewski (1979) showed that under fundamental risk it can take noise traders a very long time to lose their money, so their price impact can persist for practically relevant horizons even though it eventually vanishes. Shiller (1984) and Campbell and Kyle (1987) showed that aversion to ordinary fundamental risk can by itself sharply limit arbitrage. What was missing was a demonstration that arbitrage can be limited, and noise traders' relative wealth need not decline, even in a world purged of fundamental risk altogether — that is, a demonstration that the unpredictability of irrationality is itself a

distinct and sufficient source of risk. That is the paper's contribution, and the review below evaluates how convincingly the model establishes it, what it does and does not imply for the specific empirical anomalies the authors invoke, and where the argument is open to challenge.

2 Research Problem and Hypothesis

2.1 Research Question

The paper asks whether the unpredictability of irrational investors' future beliefs — as distinct from the riskiness of the asset's underlying cash flows — can by itself constitute a priced risk that limits arbitrage enough to (i) sustain a persistent gap between an asset's price and its fundamental value, and (ii) allow the irrational traders who create that gap to earn a higher expected return, not a lower one, than the rational arbitrageurs who trade against them. It is a narrower and more precise question than "do irrational traders affect prices," and the model is constructed specifically to isolate noise trader risk from every other channel that could produce the same qualitative result.

2.2 Motivation

The question matters because the standard rebuttal to behavioral critiques of market efficiency — that mispricing invites arbitrage, and arbitrage is unconstrained and eventually decisive — had never been tested against a model in which arbitrage is risky purely because of the psychology of other traders rather than the economics of the asset. If the rebuttal fails even in the friendliest possible setting for it (no fundamental risk, common knowledge of noise traders' distribution of beliefs, fully rational and utility-maximizing arbitrageurs), then it fails a fortiori once real-world frictions such as short-selling constraints, agency problems inside asset management, and genuine fundamental uncertainty are added back in. The paper is therefore best read as a proof of concept for the entire limits-to-arbitrage research program that followed it, most explicitly in Shleifer and Vishny (1997).

2.3 Existing Literature

Three strands preceded the paper. The market-selection strand (Friedman 1953; Fama 1965) argued that mispricing is self-correcting because irrational traders lose money to arbitrageurs and are competed out of the market. The fundamental-risk-limits-arbitrage strand (Figlewski 1979; Shiller 1984; Campbell and Kyle 1987) showed that ordinary risk aversion toward an asset's cash-flow risk can slow this correction, sometimes for a very long time, but did not dispense with fundamental risk as the source of the friction. A third, rational-heterogeneous-information strand (Grossman and Stiglitz 1980; Milgrom and Stokey 1982; Townsend 1983) modeled apparent mispricing as the equilibrium outcome of costly information acquisition among fully rational agents, an approach DSSW explicitly set themselves apart from because they find the idea of "private information" about aggregate stock-market fluctuations, attributed to traders such as Joe Granville who claim to follow a technical "system," strained as a description of what those traders are actually doing (Section V of the paper).

2.4 Research Gap

No existing model had shown that noise traders' effect on prices, and their relative returns, survive in the complete absence of fundamental risk. Without such a demonstration, a skeptic could always attribute any residual noise trader effect in the Figlewski/Shiller/Campbell-Kyle models to fundamental risk aversion happening to interact with irrational demand, rather than to the unpredictability of irrationality being a distinct economic force. The gap was conceptual, not empirical: a clean identification, within a fully specified equilibrium model, of a risk that is created by noise traders and would not exist in their absence.

2.5 Contribution

PAPER The authors claim three contributions: a closed-form equilibrium pricing function in which noise trader risk alone drives a wedge between price and fundamental value; a decomposition showing the conditions under which noise traders earn higher expected returns than arbitrageurs; and an extension, via an imitation-based population dynamic, showing that the noise-trader share of the market need not shrink to zero over time. **CRITIQUE** The first two contributions are rigorously established within the model as stated. The third is weaker than the first two: the imitation dynamics are appended rather than derived from optimizing behavior (new investors mechanically switch strategy in proportion to a lagged return differential), and the authors themselves show that a more psychologically defensible imitation rule — one based on realized utility rather than realized wealth — reverses the result and pushes the noise-trader share to zero. The paper's headline claim that noise traders can survive indefinitely therefore rests on which of two imitation rules the reader finds more plausible, a judgment call the authors are transparent about but that is easy to lose sight of when the result is cited.

3 Theoretical Framework and Mechanism

3.1 Environment

The model is a stripped-down two-period overlapping-generations economy (Samuelson 1958). Agents are born, choose a portfolio when young, and consume everything when old; there is no first-period consumption, no labor-supply margin, and no bequest, so the only economic decision in the model is the portfolio choice itself. Two assets exist. The safe asset s pays a fixed real dividend r and is in perfectly elastic supply, so its price is fixed at one and r is simply the riskless rate. The unsafe asset u pays the identical fixed dividend r , but its supply is fixed at one unit and cannot be created or destroyed. If the two assets' prices were determined solely by the present value of an identical dividend stream, they would be perfect substitutes selling at the same price of one in every period. The entire content of the model lies in showing why they do not.

There are two types of two-period-lived agents: sophisticated investors, in measure $1-\mu$, who have rational expectations, and noise traders, in measure μ , who misperceive the expected price of the risky asset next period by an i.i.d. normal random variable:

$$\rho_t \sim N(\rho^*, \sigma_\rho^2) \quad (1)$$

- ρ^* the mean misperception — a measure of noise traders' average "bullishness."
 σ_ρ^2 the variance of noise traders' period-to-period misperception of the risky asset's expected return.
 μ the fixed population share of noise traders among each generation of young agents.

Every agent has constant absolute risk aversion over wealth when old, $U(w) = -\exp(-2\gamma w)$, which under normally distributed returns reduces to maximizing the mean-variance certainty equivalent

$$E[w] - \gamma \cdot \text{Var}[w] \quad (2)-(3)$$

where γ is the common coefficient of absolute risk aversion. The sophisticated investor's objective, given his correct assessment of the distribution of next period's price, is

$$\max_{x_t^i} x_t^i \cdot [r + E_\mu p_{t+1} - (1+r)p_t] - \gamma (x_t^i)^2 \sigma_{t,p+1}^2 \quad (4)$$

and the noise trader's objective is identical except that his perceived excess return is inflated by his misperception ρ_t :

$$\max_{x_t^n} x_t^n \cdot [r + E_\mu p_{t+1} - (1+r)p_t + \rho_t] - \gamma (x_t^n)^2 \sigma_{t,p+1}^2 \quad (6)$$

where $\sigma_{t,p+1}^2$ denotes the one-period-ahead variance of the price of u , common to both agent types because it is a property of the market, not of any individual's beliefs. The first-order conditions of (4) and (6) give each type's demand for the risky asset:

$$x_t^i = \frac{r + E_\mu p_{t+1} - (1+r)p_t}{2\gamma \sigma_{t,p+1}^2} \quad (7)$$

$$x_t^n = \frac{r + E_\mu p_{t+1} - (1+r)p_t + \rho_t}{2\gamma \sigma_{t,p+1}^2} \quad (8)$$

Both types' demand is proportional to the perceived excess return and inversely proportional to the perceived one-period-ahead price variance. This last feature is what makes the model work: because next period's price depends on next period's young noise traders' as-yet-unrealized misperception, both agent types — not just the noise traders — face genuine uncertainty about the price at which they will be able to sell. That uncertainty caps every investor's position, including the sophisticated investor's, and is the wedge that keeps arbitrage from closing the gap between price and fundamental value.

3.2 The Equilibrium Pricing Function

Because the old simply liquidate, market clearing requires that the young's aggregate demand for the fixed unit supply of u equal one: $(1-\mu)x_t^j + \mu x_t^n = 1$. Substituting (7) and (8) gives

$$p_t(1+r) = r + E p_{t+1} - 2\gamma\sigma_{t,p+1}^2 + \mu\rho_t \quad (9)$$

Imposing the steady-state requirement that the unconditional distribution of p_{t+1} equal that of p_t allows (9) to be solved recursively into a closed form that depends only on the model's structural parameters and on the current realization of noise traders' misperception:

$$p_t = 1 + \frac{\rho_t - \rho^*}{1+r} + \frac{\rho^*}{r} - \frac{2\gamma\mu^2\sigma_\rho^2}{r(1+r)^2} \quad (12)$$

Equation (12) is the paper's central result, and every subsequent implication follows from reading its three deviation terms:

- **Transitory noise term**, $(\rho_t - \rho^*)/(1+r)$. This term has mean zero and vanishes only when noise traders hold exactly their average belief. It is the source of excess volatility: the price of u moves with the swings in noise traders' opinion even though nothing about u 's fundamentals has changed.
- **Price-pressure term**, ρ^*/r . If noise traders are bullish on average ($\rho^* > 0$), the asset trades at a permanent premium to its riskless substitute, because sophisticated investors, bearing a smaller share of the (already reduced) risk, require a lower expected return to hold it.
- **Noise trader risk discount**, $-2\gamma\mu^2\sigma_\rho^2/[r(1+r)^2]$. This term is always negative and does not require noise traders to be biased in any direction — it is present even at $\rho^*=0$, driven purely by the variance σ_ρ^2 of their opinions. This is the term the paper's title refers to: sophisticated investors will not hold the risky asset at its fundamental price of one unless compensated for bearing the risk that noise traders become more pessimistic before the investor can sell. Because this risk exists solely on account of noise traders' own unpredictability, and would not exist in their absence, the paper describes noise traders as "creating their own space."

The result that survives every robustness check in the paper is this last term. It shows that a fundamentally riskless asset trades, in expectation, strictly below its fundamental value of one whenever $\mu > 0$ and $\sigma_\rho^2 > 0$, regardless of the sign or size of ρ^* . **INFERENCE** Because both noise traders and sophisticated investors agree, in expectation, that u is worth one, and both would in principle like to trade toward that belief, the fact that the price nonetheless sits below one is a pure equilibrium consequence of the fact that neither type can be certain what the other type will believe next period, not evidence that either type is failing to optimize given its own beliefs.

3.3 Relative Expected Returns: Four Offsetting Effects

The paper's second major result addresses Friedman's market-selection claim directly: does the noise trader's mispricing of u cost him money relative to the sophisticated investor? The demand gap between the two types, obtained by differencing (7) and (8) and substituting the steady-state price variance, is $x_t^n - x_t^i = \rho_t(1+r)^2/(2\gamma\mu^2\sigma_\rho^2)$ (eq. 15), and the expected excess return on u consistent with market clearing is $r + E_t p_{t+1} - (1+r)p_t = 2\gamma\sigma_{t,p+1}^2 - \mu\rho_t$ (eq. 16, a rearrangement of eq. 9). Multiplying the two and taking the period- t conditional expectation gives

$$E_t[\Delta R^{n-i}] = \rho_t - \frac{(1+r)^2\rho_t^2}{2\gamma\mu\sigma_\rho^2} \quad (17)$$

and taking the unconditional expectation over the distribution of ρ_t in (1), using $E[\rho_t] = \rho^*$ and $E[\rho_t^2] = \rho^{*2} + \sigma_\rho^2$, gives the paper's headline expression for the average return gap:

$$E[\Delta R^{n-i}] = \rho^* - \frac{(1+r)^2[\rho^{*2} + \sigma_\rho^2]}{2\gamma\mu\sigma_\rho^2} \quad (18)$$

The paper reads (18) as the sum of four named effects, and this decomposition is the paper's most cited analytical device:

Table 1. The four effects decomposing the noise trader – sophisticated investor expected-return gap

Effect	Source in equation (18)	Sign	Economic content
Hold more	the leading term ρ^*	Raises ΔR when $\rho^* > 0$	Bullish noise traders hold more of the risky asset on average and so capture a larger share of the reward to risk-bearing.
Price pressure	$-(1+r)^2\rho^{*2}$ in the numerator	Always lowers ΔR , quadratically in ρ^*	Bullish demand itself bids up the price of u , compressing the very risk premium noise traders are trying to capture.
Friedman (buy-high, sell-low)	$-(1+r)^2\sigma_\rho^2$ in the numerator	Always lowers ΔR	Because misperceptions are i.i.d., noise traders have the worst possible market timing on average: they buy most heavily exactly when other noise traders are also buying, i.e., near local price peaks.
Create space	the denominator $2\gamma\mu\sigma_\rho^2$	Raises ΔR as σ_ρ^2 or γ rises	The more volatile noise traders' beliefs, or the more risk averse arbitrageurs, the more compensation sophisticated investors demand to bear noise trader risk, which softens the price-pressure and Friedman penalties.

Source: constructed by the reviewer from the algebraic decomposition and the accompanying prose in De Long et al. (1990, sec. II.A, pp. 713–15).

Two boundary results follow immediately and are the paper's sharpest empirical predictions about who profits from noise trading. If $\rho^* \leq 0$, the hold-more effect vanishes or reverses and (18) is unambiguously negative: noise traders cannot out-earn sophisticated investors unless they are bullish on average. And as ρ^* grows large for fixed σ_ρ^2 , the price-pressure term, which grows with ρ^{*2} , eventually dominates the linear hold-more term, so (18) again turns negative. **INFERENCE** Noise traders earn a higher expected return than arbitrageurs only for an *intermediate* range of average bullishness — a genuinely falsifiable, non-monotonic prediction that distinguishes the model from a naive story in which "more bullish noise traders always do better."

3.4 Population Dynamics: Does the Noise-Trader Share Survive?

Sections I–II establish that noise traders can, for some parameters, out-earn arbitrageurs within a single generation. Section III asks whether that is enough for noise trading to persist across generations, using an imitation rule in place of an optimizing entry decision: a fraction of each new generation, proportional to the lagged realized return gap, switches into whichever strategy just did better,

$$\mu_{t+1} = \max\{0, \min[1, \mu_t + \varphi(R_t^n - R_t^i)]\} \quad (20)$$

where φ governs how sensitive imitation is to the realized gap. In the pure model without fundamental risk, the paper defines a threshold noise-trader share μ^* (eq. 23) such that μ_t tends to shrink whenever it is below μ^* and tends to grow whenever it is above μ^* : the dynamics have an unstable interior threshold, so in the long run the noise-trader share is driven either toward zero or toward domination of the market, illustrated in the paper's figure 1. Adding fundamental dividend risk, u paying $r + \varepsilon_t$ rather than the certain r , changes this qualitatively. With fundamental risk present, the demand functions and pricing rule (27) replace pure noise trader risk with total risk — the sum of noise-trader-generated price risk and fundamental dividend risk — in their denominators, and the return gap converges, as $\mu \rightarrow 0$, not to negative infinity but to

$$\lim_{\mu \rightarrow 0} E[\Delta R^{n-i}] = \rho^* \quad (29)$$

a strictly positive limit whenever $\rho^* > 0$. The economic reason is that with fundamental risk present, both types' demands stay bounded as $\mu \rightarrow 0$ — a vanishingly small number of noise traders cannot absorb the entire supply of u , so sophisticated investors must hold positive amounts of the risky asset in equilibrium and must therefore be compensated with a positive risk premium, part of which noise traders continue to earn. This yields the paper's formal Proposition (reproduced from the text, proof available from the authors on request): for a sequence of economies in which the imitation-sensitivity parameter $\varphi_n \rightarrow 0$, there exists $\delta > 0$ such that the expected steady-state noise-trader share $E(\mu) \geq \delta$ for all n . In words: however slowly new investors update their strategy choice, the noise-trader population share cannot be driven arbitrarily close to zero once fundamental risk is present, because the compensation noise traders earn from the risk-premium floor in (29) is bounded away from zero.

CRITIQUE This robustness is conditional on the specific imitation rule (20). The paper is explicit that an alternative, and arguably more defensible, rule in which conversion responds to the realized *utility* gap rather than the realized *return* gap reverses the conclusion: because sophisticated investors maximize true expected utility while noise traders bear extra variance for their extra return, the noise-trader share converges to zero under utility-based imitation regardless of fundamental risk. The authors prefer the return-based rule on the empirical grounds that investors observed in practice appear to imitate strategies based on realized performance without correcting for the risk taken to achieve it, and because Friedman's original argument was itself phrased in terms of realized returns rather than realized utility. That is a defensible modeling choice, but it means the paper's most quoted claim — that noise traders need not disappear — holds only under one of two economically plausible imitation rules, not as a generic property of the model.

4 Application to Documented Market Anomalies

The paper contains no data, sample, or estimation of its own. This section is therefore an adaptation, per the review template's guidance for theoretical papers, of the "data and empirical strategy" and "results" sections into an assessment of how the closed-form results of Section 3 map onto empirical regularities that the authors cite from the existing literature, and how tight that mapping actually is.

4.1 Excess Volatility and Mean Reversion

PAPER Because the transitory term in (12) has variance $\sigma_\rho^2/(1+r)^2$ that is entirely unrelated to any change in fundamentals, the model produces volatility with no fundamental counterpart by construction. The authors invoke Shiller's (1981) variance-bound evidence (while noting Kleidon's 1986 statistical critique of it), Roll's (1984) orange-juice futures study, and Campbell and Kyle's (1987) decomposition of market movements as independent, non-model-dependent evidence that a meaningful share of observed price variation cannot be tied to news about fundamentals. If noise traders' misperceptions follow a persistent, mean-reverting process such as AR(1) with autoregressive parameter φ , the unconditional price variance

$$Var(p) = \frac{\sigma_\rho^2}{[r+(1-\varphi)]^2(1-\varphi^2)} \quad (31)$$

grows sharply as $\varphi \rightarrow 1$, i.e., as misperceptions become more persistent, while (32) shows that noise traders can still out-earn arbitrageurs even when this persistence generates very large deviations from fundamental value. The authors cite Fama and French (1988b) and Poterba and Summers (1988) for direct evidence of negative serial correlation in long-horizon returns, and Campbell and Shiller (1987) and Fama and French (1988a) for the predictive power of dividend-price and earnings-price ratios, both consistent with the mean-reverting transitory component the model implies.

CRITIQUE The mapping here is qualitative, not quantitative: the paper does not calibrate σ_ρ^2 , φ , or μ to any data series, so it cannot say whether the amount of excess volatility the model can generate for economically plausible risk-aversion and noise-trader-share parameters is of the right order of magnitude to match, say, Shiller's variance-bound violations. The claim is an existence result — noise trader risk *can* generate excess volatility and mean reversion — not a quantitative account of how much of the observed volatility it explains.

4.2 The Closed-End Fund Discount

PAPER This is the paper's most concrete and most influential application. A closed-end fund's net asset value is directly observable (the market value of its underlying portfolio), unlike the fundamental value of an ordinary stock, so the fund's price relative to net asset value is a rare direct test of whether price tracks fundamentals. Closed-end funds nonetheless trade at large, variable discounts (Malkiel 1977; Herzfeld 1980), narrower in bull markets and wider in bear markets. Reinterpreting the safe asset s as the underlying portfolio and the unsafe asset u as the fund itself, the model implies a persistent average discount driven purely by noise trader risk on the fund's own price — no average bearishness ($\rho^*=0$ is sufficient) — provided that risk is systematic across funds rather than idiosyncratic, which the authors support with Herzfeld's (1980) observation that discounts co-move across funds. The theory further predicts, and the authors argue existing agency-cost and unrealized-capital-gains explanations cannot: (i) new funds are launched exactly when noise traders are unusually optimistic, i.e., when existing funds trade at a premium; (ii) discounts should narrow sharply on announcement of open-ending or liquidation, consistent with Brauer (1984) and Brickley and Schallheim (1985), because a fund conversion removes the noise trader risk on the wrapper itself; and (iii) discounts fluctuate and occasionally turn into premiums, which a fixed agency cost or a fixed tax liability cannot generate but a stochastic ρ_t does mechanically.

EXTENSION / LITERATURE This application became the empirical centerpiece of the broader noise-trader research program: Lee, Shleifer, and Thaler (1991) test the systematic-risk assumption directly, showing that closed-end fund discounts co-move with returns on small stocks, a class of security also thought to be dominated by individual (noise trader-prone) rather than institutional investors. That paper in turn drew a direct methodological challenge from Chen, Kan, and Miller (1993), who argued the correlation between fund discounts and the size premium weakens in the later part of the sample and that low- and high-institutional-ownership small stocks behave similarly, undermining the proposed sentiment channel; Chopra, Lee, Shleifer, and Thaler (1993) rebutted that the size effect itself is a noise-trader-sensitive phenomenon essentially by construction, so subsample splits by institutional ownership are not a clean test. **CRITIQUE** The exchange is illustrative of a broader problem: the closed-end fund evidence is consistent with noise trader risk, but "consistent with" is doing a lot of work, since the same co-movement pattern would also be produced by any story in which small, thinly-traded, sentiment-exposed securities share a common risk factor, whether or not that factor is behavioral in origin.

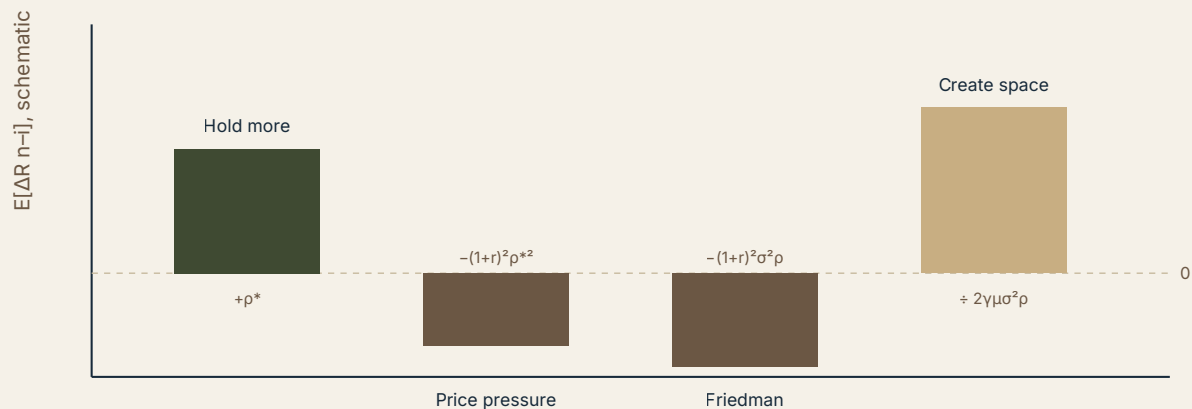
4.3 The Mehra–Prescott Equity Premium

PAPER Since noise traders can only earn a higher expected return than arbitrageurs when the risky asset is underpriced on average (eq. 33 restates the model's average price as strictly below one whenever the return gap is positive), the model offers a candidate resolution of Mehra and Prescott's (1985) puzzle that the roughly eight percent historical U.S. equity premium is far larger than a representative-agent consumption-based model can rationalize without implausibly high risk aversion. The mechanism does not operate through the standard channel of aggregate consumption covarying strongly with equity returns; instead, because noise traders' consumption-savings decisions do not satisfy the rational Euler equation, the share of wealth they invest (and hence do not consume) is low exactly when the true expected return on equities is high and high when it is low, which decouples the size of the premium from the covariance of aggregate consumption with returns. **CRITIQUE** The authors are careful to flag the logical structure of this application precisely: the model sheds light on the puzzle only if equities are in fact underpriced, which is itself the very condition required for noise traders to earn excess returns in the model. The Mehra-Prescott premium is therefore not independent confirming evidence for the model; it is, at best, indirect evidence that is equally consistent with several other equity-premium resolutions (habit formation, rare disasters, incomplete markets) that involve no noise traders at all. The paper does not attempt to adjudicate between these explanations.

4.4 Investment Horizons and Corporate Finance

PAPER The model implies that longer-lived arbitrageurs push prices closer to fundamentals, because a long-lived investor can wait through a temporary loss for the fundamental reversion, and the OLG structure is offered as a stylized representation of institutional features — frequent performance evaluation of money managers being the leading example — that impose effectively short horizons on nominally rational, well-informed capital. On corporate finance, the authors note that the standard Modigliani-Miller homemade-leverage proof only shows that a *rational* investor can undo a firm's capital-structure choice; it does not show that noise traders will do so, so if noise traders find certain, near-term cash flows easier to value correctly than uncertain, distant ones, a firm can reduce the noise trader risk borne by its securities, and hence raise their price, by paying dividends, retiring debt with equity, or repurchasing shares — predictions the paper links qualitatively to the market reactions to payout and leverage announcements catalogued in Jensen (1986).

CRITIQUE Both applications in this subsection are advanced only qualitatively; the paper explicitly states that a proper treatment of idiosyncratic, firm-specific noise trader risk would require adding transaction costs that limit diversification (citing Mayshar 1983), which lies outside the model actually built. They are best read as suggestive extrapolations of the mechanism rather than results the model formally establishes.

Figure 1. The four-effect decomposition of the noise trader – arbitrageur return gap, equation (18)

Constructed by the reviewer to illustrate the sign and role of each term in equation (18); bar heights are illustrative and not calibrated to any parameter values. Source: derived from De Long, Shleifer, Summers, and Waldmann (1990), pp. 713–15.

5 Results: What the Model Formally Establishes

Table 2. The paper's formal propositions and comparative statics

Claim	Formal status	Condition required
Price of a fundamentally riskless asset sits below its riskless substitute's price	Closed-form equilibrium result, eq. (12)	$\mu > 0$ and $\sigma^2\rho > 0$; holds regardless of ρ^* sign
Noise traders earn a higher expected return than arbitrageurs	Closed-form comparative static, eq. (18)	ρ^* strictly positive and not too large (interior range)
Noise trader risk raises sophisticated investors' expected utility	Proved directly from the enlarged opportunity set, sec. II.B	Holds unconditionally in the baseline (fixed-supply) model
Noise-trader population share bounded away from zero in the long run	Formal Proposition, proof available on request	Fundamental risk present and imitation based on realized returns, not realized utility
Longer investor horizons push price toward fundamental value	Informal argument via an infinitely lived arbitrageur example	Not derived as a formal limit result for general finite horizons $n > 2$

Source: constructed by the reviewer from the propositions and derivations in De Long, Shleifer, Summers, and Waldmann (1990).

Read against Table 2, the paper's rhetorical framing is broader than what is formally shown. The abstract and conclusion present the model as explaining "excess volatility... mean reversion... underpricing of closed-end mutual funds, and the Mehra-Prescott equity premium puzzle," but only the closed-end fund discount is derived as a direct reinterpretation of the two-asset structure; the other three are qualitative extrapolations supported by citations to independent empirical

papers rather than formal results of the model as specified. This is not a flaw in the paper's honesty — the authors flag the distinction themselves at several points, most explicitly around the equity-premium discussion — but it is a distinction a reader should not lose when the paper is cited as having "explained" all four anomalies.

6 Robustness and Additional Tests

CONCERN → TEST → RESULT

Concern: the core result might be an artifact of the overlapping-generations structure itself rather than of noise trader risk. **Test:** the authors verify that equilibrium exists whenever the return to holding the risky asset is always uncertain, which the OLG structure guarantees via the absence of a terminal period; they show that a model with a known terminal liquidation date and no fundamental risk has *no* equilibrium at all, because both types would then attempt unbounded riskless arbitrage. **Does it resolve the concern:** partially. It shows the OLG assumption is not doing independent work as long as some form of terminal uncertainty exists, but it also reveals that the model is fragile to the removal of that uncertainty altogether — equilibrium existence, not just the size of the noise trader discount, depends on it.

CONCERN → TEST → RESULT

Concern: the specific functional form assumed for noise trader demand — misperceiving the expected *price* by an additive normal shock — might be driving the result. **Test:** a footnote states that the equilibrium in which noise traders matter survives under two alternative primitives: noise traders fixing a price at which they trade unlimited quantities, or noise traders misperceiving the *variance* rather than the mean of returns. **Does it resolve the concern:** the claim is asserted rather than shown in the published text (the derivations are relegated to an unpublished appendix "available from the authors on request"), so a reader cannot verify the robustness claim from the paper itself.

CONCERN → TEST → RESULT

Concern: the survival of noise traders in Section III might depend on the specific returns-based imitation rule chosen. **Test:** the authors solve the model under an alternative utility-based imitation rule. **Does it resolve the concern:** no — the alternative rule reverses the conclusion, driving the noise-trader share to zero. The authors argue for preferring the returns-based rule on empirical plausibility grounds (investors imitate realized performance, not risk-adjusted performance) rather than showing the result is robust across specifications; this is a defensible but explicit concession that the persistence result is rule-dependent, not structural.

CONCERN → TEST → RESULT

Concern: the model assumes returns can be negative, which is economically odd for a claim about "wealth." **Test:** the authors acknowledge that unbounded normal returns give every investor positive probability of negative final wealth and consumption, and state that an appendix to the 1987 NBER working paper version presents an example with strictly positive prices and consumption throughout. **Does it resolve the concern:** the published paper does not reproduce that example, so the claim that the qualitative results survive a strictly-positive-consumption respecification cannot be checked directly from the 1990 text.

7 Critical Evaluation

7.1 What the Paper Establishes

ESTABLISHED

That noise trader risk, defined as the risk arising purely from the unpredictability of noise traders' future misperceptions, is sufficient on its own — without any fundamental risk — to (i) support an equilibrium price for a riskless asset strictly below its fundamental value, and (ii) allow noise traders to earn a higher expected return than arbitrageurs for an interior range of average bullishness. These are closed-form results, fully derived within a well-specified equilibrium model, and they directly refute the strong form of Friedman's claim that irrational traders must, as a matter of market logic, earn lower returns than the arbitrageurs who trade against them.

SUPPORTED BUT CONDITIONAL

That the noise-trader population share need not converge to zero over time. This holds only under the returns-based imitation rule and only once fundamental risk is reintroduced; it does not hold under the utility-based imitation rule the authors themselves present as an alternative, nor in the pure noise-trader-risk-only version of the model, where the population share is driven to one of two corners.

SUGGESTIVE

That the mechanism explains excess volatility, long-horizon mean reversion, the failure of the expectations hypothesis of the term structure, and the Mehra-Prescott equity premium. In each case the model shows the mechanism *can* produce the qualitative pattern; none of these applications is calibrated, and in each case the cited empirical evidence for the underlying pattern (excess volatility, mean reversion) comes from separate papers using separate identification strategies, not from a test of the noise-trader-risk model against data.

NOT ESTABLISHED

Any claim about the empirical *magnitude* of noise trader risk in actual markets, the size of μ , ρ^* , or σ_ρ^2 for any real asset class, or a structural test that could discriminate the noise-trader-risk mechanism from the rational-heterogeneous-information alternative (Grossman and Stiglitz 1980) using the same reduced-form evidence. The paper is explicit that it has not attempted such a test and regards the choice between the two families of explanation as, in part, a matter of which the reader finds more introspectively plausible (Section V).

7.2 Strengths

STRENGTHS RECOGNIZED BY THE AUTHORS The authors emphasize the model's tractability and its ability to isolate noise trader risk by construction, since the two assets are engineered to be identical in every fundamental respect. **STRENGTHS IDENTIFIED INDEPENDENTLY** Three further strengths stand out on inspection. First, the closed-form pricing function (12) is unusually transparent for a model of its type: every term has a clean economic label, which is precisely what has made the four-effect decomposition in Table 1 so durable as a teaching and citation device across three decades of subsequent literature. Second, the paper is scrupulous about flagging the limits of its own applications — the equity-premium discussion in particular is a model of appropriate epistemic hedging, explicitly noting the circularity between the model's premise and the evidence it is meant to explain. Third, the decision to build the model with literally zero fundamental risk, rather than to add noise trader risk on top of fundamental risk, is a genuinely well-chosen identification device: it removes the single most obvious alternative explanation for any result the model produces.

7.3 Weaknesses and Critique

PROBLEM → WHY IT MATTERS → POSSIBLE TEST

Exogenous, unmodeled noise traders. Noise traders' beliefs are a free parameter of the model — a normal random variable with mean ρ^* and variance σ_ρ^2 — with no account of why noise traders believe what they believe, why their misperceptions would be normally distributed and serially independent (or AR(1) in the extension), or why μ is fixed rather than itself an equilibrium object in the baseline model. This matters because every one of the paper's comparative statics is conditional on these assumed moments, and the paper offers no independent measurement of them. *Possible test:* structurally estimate ρ^* and σ_ρ^2 from a directly observed sentiment proxy (for example, survey-based sentiment indices or the Baker-Wurgler 2006 sentiment index) and check whether the implied noise trader risk premium matches realized closed-end fund discounts out of sample.

PROBLEM → WHY IT MATTERS → POSSIBLE TEST

Observational equivalence with rational heterogeneous-information models. Every qualitative prediction in Section IV — excess volatility, mean reversion, a risk premium unrelated to consumption covariance — can also be generated by models with fully rational, privately informed agents (Grossman and Stiglitz 1980; Townsend's 1983 forecasting-the-forecasts-of-others framework; Stein 1987). The paper acknowledges this directly but does not propose a discriminating test. *Possible test:* exploit episodes in which a sentiment proxy moves for reasons plausibly unrelated to private fundamental information (for example, retail brokerage account openings, or search-engine attention shocks) to see whether price movements track the sentiment proxy in a way a private-information model would not predict.

PROBLEM → WHY IT MATTERS → POSSIBLE TEST

Imitation-rule sensitivity of the survival result. As discussed in Sections 3.4 and 6, the headline claim that noise traders need not be selected out of the market flips sign under an equally defensible alternative behavioral rule for how new capital chooses a strategy. This matters because the survival result is frequently cited as if it were a structural property of noise trader risk, when it is in fact a property of one particular, unmicrofounded law of motion for capital allocation. *Possible test:* estimate, from mutual-fund or hedge-fund flow data, whether capital reallocation across strategies responds more strongly to realized raw returns or to realized risk-adjusted returns, which would discipline the choice between the paper's two imitation rules empirically rather than by introspection.

7.4 The Single Most Important Weakness

The paper's central limitation is that it is, by its own design, a demonstration of possibility rather than of magnitude or of discrimination. It shows that a mechanism exists that can, in principle, generate each of the phenomena it discusses, without providing a way to measure how much of any observed anomaly the mechanism actually accounts for, or a test capable of distinguishing its predictions from those of a rational model with heterogeneous information. This is not a defect relative to the paper's own stated ambitions — it is explicitly a theoretical contribution — but it is the reason the empirical literature it inspired (Lee, Shleifer, and Thaler 1991; the Chen-Kan-Miller exchange; Shleifer and Vishny 1997) had to do a great deal of additional work before the noise-trader-risk mechanism could be treated as more than a logical possibility.

8 Contradictions and Edge Cases

What happens as $\mu \rightarrow 0$ without fundamental risk? Equation (18)'s denominator shrinks with μ , so the expected return gap diverges to negative infinity: an infinitesimally small population of noise traders, in the pure model, loses an arbitrarily large amount relative to arbitrageurs, because the two groups' demands become unbounded as they each perceive an almost riskless arbitrage against a shrinking counterparty. **INFERENCE** This is an economically sensible limit — it recovers

Friedman's intuition exactly in the boundary case where noise traders are too few to generate meaningful noise trader risk — but it also shows that the model's headline result depends on μ being non-trivial, not merely positive; the paper does not say how large μ must be for the mechanism to matter quantitatively.

What happens as $\mu \rightarrow 0$ with fundamental risk? The same limit instead converges to the finite, strictly positive value ρ^* (eq. 29), because fundamental risk keeps both types' demands bounded. **CONTRADICTION CHECK** This is a genuine discontinuity in the model's comparative statics at $\sigma_\varepsilon^2 = 0$: an arbitrarily small amount of fundamental risk changes the $\mu \rightarrow 0$ limit from $-\infty$ to a strictly positive constant. A discontinuous comparative static of this kind is a signal that the "no fundamental risk" case is a knife-edge special case rather than a robust benchmark, and readers should treat the pure noise-trader-risk-only results in Sections I-II with some caution when extrapolating to markets that always have at least some fundamental uncertainty.

What happens as ρ^* becomes large? The price-pressure term in (18) grows with ρ^{*2} while the hold-more term grows only linearly, so for sufficiently bullish noise traders the return gap turns negative again: extremely bullish noise traders are eventually punished, not rewarded, by their own bullishness. This produces the non-monotonic, single-peaked relationship between average noise trader sentiment and noise trader profitability noted in Section 3.3, an implication that is easy to overlook when the paper is summarized as "noise traders can earn higher returns."

Does the result survive outside the two-asset, two-period setting? The paper's own extension to longer-lived agents (Section IV.D) shows the effect shrinks, not disappears, as horizons lengthen, because a longer-lived arbitrageur can wait through a temporary loss and collects intervening dividends as partial insurance. In the literal limit of an infinitely lived, risk-averse arbitrageur facing a known and finite liquidation date, the paper shows the price is driven exactly to fundamental value. **INFERENCE** The entire mechanism is therefore best understood as a statement about the interaction of short horizons and noise, not about noise trader risk as an economy-wide constant; any institutional change that lengthens the effective horizon of the marginal arbitrageur (removing frequent performance evaluation, for example) should, within the logic of the model, shrink noise trader risk's price impact.

Could another mechanism generate the same closed-end fund pattern? A tax-timing story or an agency-cost story could each generate a persistent average discount, though the paper argues neither easily generates a fluctuating discount that occasionally turns into a premium, nor the co-movement of discounts across unrelated funds. A liquidity-based explanation (funds are relatively illiquid, and illiquidity risk is priced) was not part of the 1990 comparison set and remains, in the reviewer's assessment, a live alternative that the paper's own evidence does not rule out, since illiquidity, like noise trader sentiment, would plausibly also co-move across small, thinly traded closed-end vehicles.

9 Further Literature

PRECEDED IT

Friedman (1953), "The Case for Flexible Exchange Rates." Shared idea: destabilizing speculation is unprofitable and self-eliminating. Relationship: the paper this article is written to refute; DSSW's entire model is constructed to show the argument's implicit riskless-arbitrage assumption fails. Why it matters: without Friedman's claim as the target, the paper's central results have no rhetorical or theoretical point of reference.

Fama (1965), "The Behavior of Stock Market Prices." Shared idea: rational arbitrageurs compete away the price impact of uninformed trading. Relationship: the empirical companion to Friedman's theoretical claim; DSSW's model implies this need not hold once arbitrage is risky. Why it matters: situates the paper squarely against the empirical efficient-markets tradition, not only its theoretical foundations.

Figlewski (1979), "Subjective Information and Market Efficiency in a Betting Market." Shared idea: noise traders can retain price influence for a long time. Relationship: an earlier, narrower demonstration using fundamental risk; DSSW generalize the mechanism by removing fundamental risk entirely. Why it matters: shows DSSW's contribution is a strengthening, not a discovery, of the basic insight that arbitrage can be slow.

Shiller (1984) and Campbell and Kyle (1987, unpublished manuscript). Shared idea: risk aversion toward fundamental risk limits arbitrage. Relationship: DSSW's direct predecessors in the limits-to-arbitrage tradition; DSSW's innovation is to show the same qualitative limitation survives even without any fundamental risk to be averse to. Why it matters: clarifies exactly what is new in DSSW relative to the immediately preceding literature.

Black (1986), "Noise." Shared idea: coins the term "noise trader" and argues noise is what makes markets liquid and observation of true value impossible in real time. Relationship: DSSW build directly on Black's terminology and diagnosis, formalizing it into an equilibrium asset-pricing model. Why it matters: DSSW is best read as the first rigorous general-equilibrium formalization of Black's informal argument.

Kyle (1985), "Continuous Auctions and Insider Trading." Shared idea: models noise traders (there, "liquidity traders") as a necessary counterparty that allows informed trading to occur without immediately revealing information. Relationship: a different market-microstructure use of noise trading, cited by DSSW via Black (1986) as the origin of the term in the finance literature. Why it matters: distinguishes Kyle's microstructure notion of noise (liquidity demand as camouflage) from DSSW's macro-asset-pricing notion of noise (systematic, priced sentiment risk); conflating the two is a common misreading in secondary literature.

Grossman and Stiglitz (1980), "On the Impossibility of Informationally Efficient Markets." Shared idea: apparent mispricing and trading volume can arise among fully rational agents. Relationship: the paper's explicitly acknowledged rival explanation; DSSW state they find the "private

information" interpretation of noise trader behavior implausible for aggregate market fluctuations but do not construct a test that discriminates the two frameworks. Why it matters: this is the central identification problem the paper leaves unresolved, discussed further in Section 8 above.

CONTEMPORANEOUS COMPANION PAPERS BY THE SAME AUTHORS

De Long, Shleifer, Summers, and Waldmann (1989), "The Size and Incidence of the Losses from Noise Trading," *Journal of Finance* 44. Extends the wealth-dynamics question the 1990 paper leaves informal, examining how large noise traders' losses can be under alternative assumptions. **De Long, Shleifer, Summers, and Waldmann (1990), "Positive Feedback Investment Strategies and Destabilizing Rational Speculation," *Journal of Finance* 45.** Shows that once rational traders can anticipate future noise trader demand shifts rather than only bet on mean reversion, their trading can itself become destabilizing, a direct extension of the anticipatory-strategy possibility flagged but not modeled in Section IV.A of the reviewed paper.

APPEARED LATER AND DEVELOPED ITS CONTRIBUTION

Lee, Shleifer, and Thaler (1991), "Investor Sentiment and the Closed-End Fund Puzzle," *Journal of Finance* 46. Shared idea: closed-end fund discounts as a measurable proxy for individual investor sentiment. Relationship: the direct empirical successor that turns DSSW's closed-end fund application into a testable, general sentiment index and links it to the size effect. Why it matters: converted the reviewed paper's most concrete application into an empirically operational research program, at the cost of the Chen-Kan-Miller (1993) methodological challenge discussed in Section 4.2.

Shleifer and Vishny (1997), "The Limits of Arbitrage," *Journal of Finance* 52. Shared idea: arbitrage is capital-constrained and risky, so mispricing can persist. Relationship: generalizes DSSW's mechanism by adding agency frictions between arbitrageurs and the outside investors who fund them, showing mispricing can worsen, not just persist, precisely when arbitrage capital is most needed. Why it matters: the field's standard reference for "limits to arbitrage" today builds its performance-based-outflow mechanism on the same short-horizon logic DSSW formalized first.

Barberis, Shleifer, and Vishny (1998), "A Model of Investor Sentiment," *Journal of Financial Economics* 49. Shared idea: systematic, psychologically motivated belief errors that affect prices. Relationship: moves from DSSW's reduced-form normal-shock noise trader to a structural model of investor under- and over-reaction grounded in specific cognitive biases. Why it matters: represents the next methodological step the reviewed paper's conclusion explicitly calls for — "more primitive descriptions of irrationality" that generate sharper, more falsifiable predictions.

Baker and Wurgler (2006), "Investor Sentiment and the Cross-Section of Stock Returns," *Journal of Finance* 61. Shared idea: a directly measured, market-wide sentiment index. Relationship: operationalizes the previously unobserved ρ_t of the DSSW model as an empirically constructed composite index and tests its cross-sectional pricing implications. Why it matters: the clearest instance of DSSW's abstract "noise trader misperception" parameter being turned into a measurable, tradable object thirty years after the reviewed paper's publication.

10 Research Gaps

PAPER-LEVEL GAPS

The paper does not explain why noise traders hold the beliefs they hold, does not endogenize the population share μ in the baseline model, and does not calibrate any of its parameters to real markets. It also leaves the promised robustness to alternative noise trader demand specifications (footnote 3) unverified in the published text, and does not reproduce the strictly-positive-consumption respecification mentioned as available in the unpublished 1987 working paper appendix.

LITERATURE-LEVEL GAPS

Thirty-six years on, the discrimination problem between rational-heterogeneous-information models (Grossman-Stiglitz) and noise-trader-risk models remains largely unresolved for aggregate market-level phenomena, even though sentiment measures such as Baker-Wurgler (2006) have made the noise trader channel more directly testable at the cross-sectional level. There is still no widely accepted structural estimate of how much of observed equity volatility is attributable to noise trader risk specifically, as opposed to time-varying rational risk premia.

METHODOLOGICAL GAPS

Better identification would require data that isolates variation in investor sentiment that is plausibly unrelated to private fundamental information (a requirement that survey-based and attention-based sentiment proxies only partially satisfy), better measurement of the effective horizon of the marginal arbitrageur in a given asset class (a quantity the model shows to be central but that is rarely observed directly), and estimation methods robust to the fact that noise trader risk, if it exists, would itself generate the fat-tailed, volatility-clustered return data that complicate standard inference.

GAP → WHY IT MATTERS → WAY FORWARD

Gap: no discriminating test between noise-trader-risk and rational-private-information explanations of the same reduced-form patterns. **Why it matters:** the entire behavioral-finance research program's welfare and policy implications (for example, the case for circuit breakers, transaction taxes, or sentiment-based regulatory intervention) depend on prices being wrong in a way individual rationality alone would not correct. **Way forward:** exploit natural experiments that shift measured sentiment for reasons orthogonal to private information about fundamentals, and test whether price responses match the noise-trader-risk model's specific comparative statics (for example, the non-monotonic relationship between average sentiment and noise trader profitability in eq. 18) rather than only its qualitative sign predictions, which a private-information model can typically also match.

11 Research Opportunities

1. A structural, non-monotonic test of the noise-trader return-gap prediction using retail brokerage sentiment

RESEARCH QUESTION: Does the empirical relationship between measured retail-investor sentiment and retail investors' realized portfolio returns relative to institutional benchmarks exhibit the interior-maximum, non-monotonic shape implied by equation (18)?

MOTIVATION: The paper's sharpest and least-tested prediction — that noise traders out-earn arbitrageurs only over an intermediate range of average bullishness, not monotonically — has never, to the reviewer's knowledge, been tested directly against realized retail trading records.

HYPOTHESIS: Retail portfolio return differentials relative to a market benchmark, sorted by contemporaneous retail sentiment quintile, will rise and then fall as sentiment moves from bearish to strongly bullish, tracing the single-peaked shape of eq. (18).

ECONOMIC MECHANISM: The price-pressure and Friedman effects dominate the hold-more effect once average bullishness passes a threshold set by the population share and risk-aversion parameters.

LITERATURE CONNECTION: Extends Lee, Shleifer, and Thaler (1991) and Baker and Wurgler (2006) from a co-movement test to a direct shape test of the DSSW return-gap function.

DATA: Retail brokerage account-level trading and holdings data (for example, through a discount broker research partnership), matched to a contemporaneous survey- or search-based sentiment index at monthly frequency.

VARIABLES: Retail portfolio excess return relative to a matched institutional or market benchmark; sentiment level and its innovation; retail equity ownership share as a proxy for μ .

METHOD: Nonparametric or quadratic-in-sentiment regression of the realized retail-institutional return gap on sentiment level, testing for a statistically significant interior maximum against the monotonic alternative.

IDENTIFICATION STRATEGY: Within-account fixed effects and instrumenting sentiment innovations with plausibly exogenous shocks (for example, unrelated macro news salience) to separate sentiment-driven from information-driven retail trading.

EXPECTED CONTRIBUTION: The first direct empirical test of the model's specific functional-form prediction rather than only its qualitative sign.

MAIN RISK: Retail brokerage data with sufficiently long history and sentiment coverage may not be accessible, and any observed non-monotonicity would remain open to a rational-learning reinterpretation.

2. Structural estimation of μ , ρ^* , and $\sigma^2\rho$ from closed-end fund discount time series

RESEARCH QUESTION: What values of the model's three free parameters best rationalize the observed time series of closed-end fund discounts, and are they economically plausible?

MOTIVATION: The paper's closed-end fund application (Section 4.2) is qualitative; no published paper structurally estimates the DSSW model's own parameters against this application.

HYPOTHESIS: A method-of-moments or simulated-minimum-distance estimation of (12) restricted to the fund/portfolio reinterpretation can match the mean, variance, and autocorrelation of discounts with parameter values consistent with independently measured retail ownership shares of closed-end funds.

ECONOMIC MECHANISM: The noise trader risk discount term in (12) maps directly and uniquely onto the fund discount once s and u are reinterpreted as portfolio and wrapper.

LITERATURE CONNECTION: Directly extends Lee, Shleifer, and Thaler (1991) from a reduced-form co-movement test to a fully structural estimation of the originating model.

DATA: CRSP/Compustat closed-end fund price and net-asset-value history; fund-level retail versus institutional ownership from 13F and fund-level ownership disclosures.

VARIABLES: Fund discount level and volatility; fund launch and open-ending dates; ownership concentration by investor type.

METHOD: Simulated method of moments matching model-implied discount moments to sample moments, with standard errors via block bootstrap given serial dependence.

IDENTIFICATION STRATEGY: Cross-sectional variation in retail ownership share across funds used to separately identify μ from $\sigma^2\rho$, since the model predicts discount variance scales with μ^2 .

EXPECTED CONTRIBUTION: The first parameter-level, out-of-sample-testable calibration of the paper's core equilibrium object.

MAIN RISK: Identification of three parameters from what is effectively a single time series per fund is likely to be weak without strong auxiliary restrictions or a large cross-section of funds.

3. Testing the horizon-lengthening comparative static using pension-fund evaluation-frequency reforms

RESEARCH QUESTION: Do reforms that lengthen the performance-evaluation window of institutional asset managers measurably reduce the price impact of retail sentiment on the securities those managers hold?

MOTIVATION: Section 4.4 and 3.4 identify investor horizon as the model's central, but least directly tested, comparative static.

HYPOTHESIS: Following a regulatory or plan-level shift from quarterly to annual (or longer) manager evaluation, the sensitivity of affected managers' portfolio holdings' returns to sentiment shocks declines.

ECONOMIC MECHANISM: Longer effective horizons reduce the resale-price risk borne by the marginal arbitrageur, per the infinitely lived-arbitrageur limit case discussed in Section IV.D of the paper.

LITERATURE CONNECTION: A direct test of the short-horizon mechanism shared with Shleifer and Vishny (1997).

DATA: Pension plan investment-policy documents and manager mandate changes; holdings-level returns from Form 13F or fund-level data; a sentiment index at the relevant frequency.

VARIABLES: Evaluation frequency before and after a reform; portfolio sensitivity (beta) to sentiment innovations; fund flows.

METHOD: Difference-in-differences comparing sentiment-beta of affected versus unaffected managers around the reform date.

IDENTIFICATION STRATEGY: Staggered adoption of evaluation-frequency reforms across pension plans or jurisdictions as a source of quasi-random variation in effective arbitrageur horizon.

EXPECTED CONTRIBUTION: Direct evidence for or against the model's most policy-relevant comparative static, with implications for the design of asset-management mandates.

MAIN RISK: Evaluation-frequency reforms are rare and may be correlated with other governance changes that independently affect sentiment sensitivity.

4. A discriminating test between noise-trader-risk and Grossman-Stiglitz private-information explanations using algorithmic sentiment shocks

RESEARCH QUESTION: Can a source of sentiment variation be found that is plausibly orthogonal to any private fundamental information, so that a price response to it can only be explained by a noise-trader-risk-type model?

MOTIVATION: Section 7.3 identifies the paper's central unresolved weakness as observational equivalence with rational private-information models; no research idea in the existing literature squarely targets it.

HYPOTHESIS: Sentiment shocks generated by socially salient but fundamentally uninformative events (for example, viral non-financial news coinciding with a ticker-symbol coincidence, or weather-driven mood effects at a metropolitan trading-desk level) move prices in the direction and with the persistence the DSSW model predicts, despite carrying no plausible private information.

ECONOMIC MECHANISM: If price moves with a sentiment proxy that is constructed to be information-free by design, a private-information model has no channel to generate the same response, while the noise-trader-risk model predicts it directly through p_t .

LITERATURE CONNECTION: Builds on Roll's (1984) orange-juice-futures identification strategy, cited in the reviewed paper, generalized to equities using modern high-frequency, geographically tagged trading data.

DATA: High-frequency equity trade and quote data merged with geographically and temporally tagged, information-free sentiment shocks (weather, sports outcomes, social-media virality metrics).

VARIABLES: Abnormal returns and reversal patterns around the identified sentiment shock; trading volume; order-flow imbalance.

METHOD: Event-study methodology with placebo events and a battery of falsification tests to rule out any private-information channel.

IDENTIFICATION STRATEGY: Selection of sentiment shocks specifically constructed to be uncorrelated with any plausible fundamental information set, following the logic of Roll (1984) and Hirshleifer-Shumway (2003) weather studies.

EXPECTED CONTRIBUTION: The closest available approximation to a clean discriminating test between the paper's mechanism and its main rational alternative.

MAIN RISK: True information-free-ness of any proposed sentiment shock is difficult to establish beyond reasonable doubt, and effect sizes from such shocks are typically small relative to statistical power requirements.

5. Extending the noise-trader-risk framework to on-chain sentiment and stablecoin/token discount dynamics

RESEARCH QUESTION: Does the DSSW noise-trader-risk mechanism explain persistent, fluctuating discounts of closed-end crypto investment vehicles (e.g., trust products holding a single underlying token) relative to their net asset value, analogous to the 1990s closed-end equity fund discount?

MOTIVATION: Publicly traded crypto trust products exhibit large, variable, and occasionally positive discounts/premiums to net asset value, a close institutional analogue to the paper's central application, with the advantage of complete, high-frequency, and public on-chain trading and holder data.

HYPOTHESIS: Discount fluctuations across multiple single-asset crypto trusts co-move with a common, directly measurable retail-sentiment proxy (search interest, social-media activity, retail exchange order-flow imbalance) more strongly than with fund-specific fundamentals.

ECONOMIC MECHANISM: Systematic noise trader risk on the wrapper, exactly as in Section 4.2, with the redemption mechanism (or its absence) playing the role of the open-ending event in Brauer (1984).

LITERATURE CONNECTION: A direct modern replication of Lee, Shleifer, and Thaler (1991) in a market with far richer investor-level data than 1990s equity markets provide.

DATA: On-chain wallet-level holding and transaction data for the trust's shares and the underlying token; publicly reported net asset value; search and social-media sentiment indices.

VARIABLES: Discount/premium level and volatility; wallet concentration and turnover as a direct, non-survey-based proxy for μ ; redemption-mechanism status.

METHOD: Panel regression of discount on sentiment proxies with fund and time fixed effects, plus an event study around redemption-mechanism changes.

IDENTIFICATION STRATEGY: Staggered introduction of redemption mechanisms across different trust products as quasi-random variation analogous to Brauer's (1984) open-ending events.

EXPECTED CONTRIBUTION: A test of the model's external validity in a market structurally similar to closed-end funds but with wallet-level rather than survey-inferred investor identification.

MAIN RISK: Regulatory and product heterogeneity across crypto trust vehicles may confound the comparison, and on-chain wallet identity does not map cleanly onto "noise trader" versus "sophisticated investor" without further assumptions.

12 Learning Framework

PREREQUISITE CONCEPTS

Overlapping-generations models (Samuelson 1958); constant absolute risk aversion utility and the mean-variance certainty-equivalent reduction under normal returns; the standard efficient-markets and market-selection arguments (Friedman 1953; Fama 1965); basic rational-expectations equilibrium solution by recursive substitution under a stationarity restriction.

CORE CONCEPTS LEARNED

Noise trader risk as a distinct risk category, separate from fundamental risk, that arises purely from the unpredictability of irrational agents' beliefs; the "creates its own space" logic by which unpredictability itself, rather than any bias, generates a priced discount; the four-effect decomposition (hold more, price pressure, Friedman, create space) of the noise trader–arbitrageur return gap; the distinction between imitation based on realized wealth and imitation based on realized utility, and why the two need not give the same long-run population dynamics.

IMPORTANT EQUATIONS

The equilibrium pricing function (12), which decomposes price deviation from fundamental value into a transitory noise term, a price-pressure term, and the always-negative noise trader risk discount; and the return-gap equation (18), which is the paper's most frequently cited single expression and the basis of Table 1's four-effect decomposition.

METHODS LEARNED

Solving a rational-expectations asset-pricing model by recursive substitution under a steady-state (stationary-distribution) restriction rather than by guessing a functional form; decomposing a comparative-static expression into named, economically interpretable additive and multiplicative components; using an exogenous imitation rule as a tractable, if unmicrofounded, substitute for a fully dynamic optimizing model of capital allocation across strategies.

QUESTIONS TO TEST UNDERSTANDING

Comprehension. In your own words, explain why the price of the risky asset u sits below its fundamental value of one even when noise traders are, on average, neither bullish nor bearish ($\rho^*=0$). What role does $\sigma^2\rho$ play that ρ^* does not?

Analysis. Why does the paper need to add fundamental dividend risk to the model before it can show that the noise-trader population share is bounded away from zero in the long run? What specifically breaks in the $\mu \rightarrow 0$ limit of the pure noise-trader-risk model that fundamental risk repairs?

Research. Design a test that could, in principle, distinguish the noise-trader-risk explanation of closed-end fund discounts from a rational, private-information-based explanation of the same discounts. What data would you need that the original 1990/1991 papers did not have access to?

PAPERS TO READ NEXT

Black (1986), "Noise" — the informal precursor whose diagnosis this paper formalizes. Lee, Shleifer, and Thaler (1991), "Investor Sentiment and the Closed-End Fund Puzzle" — the direct empirical test of the paper's central application. Shleifer and Vishny (1997), "The Limits of Arbitrage" — the field-standard generalization that adds capital and agency constraints to the same short-horizon logic.

13 Conclusion

ONE-SENTENCE THESIS

The unpredictability of irrational investors' future beliefs is, on its own and even absent any fundamental risk, a priced risk that limits arbitrage enough to sustain persistent mispricing and to let irrational traders out-earn the rational investors who trade against them.

THREE MOST IMPORTANT INSIGHTS

First, "noise creates its own space": an asset can be underpriced relative to a fundamentally identical substitute purely because neither type of investor can predict what the other type will believe next period, with no bias required in either direction. Second, the return gap between noise traders and arbitrageurs is not monotonic in noise trader bullishness; it is maximized at an interior degree of average optimism and turns negative at the extremes, because the same bullishness that raises noise traders' exposure to the risk premium also bids away that premium through price pressure. Third, whether noise traders persist in the market over time is not a structural property of the model but depends on which of two plausible imitation rules new investors follow, a fragility the authors disclose but that is frequently lost in citation.

STRONGEST EVIDENCE

The closed-form derivation of equation (12) and the return-gap decomposition in equation (18), both of which are fully derived within an explicit, transparent equilibrium model and require no auxiliary empirical assumption to hold as stated.

BIGGEST WEAKNESS

The model cannot, by construction, discriminate its noise-trader-risk mechanism from a fully rational model with privately informed, heterogeneous agents (Grossman and Stiglitz 1980); every application in Section IV of the paper is consistent with, but not uniquely implied by, the noise-trader-risk story.

MOST IMPORTANT RESEARCH GAP

The absence of any test, in the thirty-six years since publication, that cleanly discriminates noise-trader-risk explanations from rational-private-information explanations of the same reduced-form asset-pricing patterns at the aggregate market level.

BEST RESEARCH OPPORTUNITY

Research idea 4 above — constructing sentiment shocks that are plausibly information-free by design and testing whether prices respond to them in the manner the model, but not a private-information model, predicts — is, in the reviewer's assessment, the opportunity with the highest ratio of scientific value to feasibility, because modern high-frequency and alternative-data sources make the identification problem tractable in a way it was not in 1990.

THREE PAPERS TO READ NEXT

Black (1986), "Noise"; Lee, Shleifer, and Thaler (1991), "Investor Sentiment and the Closed-End Fund Puzzle"; Shleifer and Vishny (1997), "The Limits of Arbitrage."

14 References

- Baker, Malcolm, and Jeffrey Wurgler. "Investor Sentiment and the Cross-Section of Stock Returns." *Journal of Finance* 61 (2006): 1645–80.
- Barberis, Nicholas, Andrei Shleifer, and Robert Vishny. "A Model of Investor Sentiment." *Journal of Financial Economics* 49 (1998): 307–43.
- Black, Fischer. "Noise." *Journal of Finance* 41 (1986): 529–43.
- Brauer, Gregory A. "'Open-Ending' Closed-End Funds." *Journal of Financial Economics* 13 (1984): 491–507.
- Brickley, James A., and James S. Schallheim. "Lifting the Lid on Closed-End Investment Companies: A Case of Abnormal Returns." *Journal of Financial and Quantitative Analysis* 20 (1985): 107–17.
- Campbell, John Y., and Robert J. Shiller. "Cointegration and Tests of Present Value Models." *Journal of Political Economy* 95 (1987): 1062–88.
- Chen, Nai-Fu, Raymond Kan, and Merton H. Miller. "Are the Discounts on Closed-End Funds a Sentiment Index?" *Journal of Finance* 48 (1993): 795–800.
- Chopra, Navin, Charles M. C. Lee, Andrei Shleifer, and Richard H. Thaler. "Yes, Discounts on Closed-End Funds Are a Sentiment Index." *Journal of Finance* 48 (1993): 801–8.
- De Long, J. Bradford, Andrei Shleifer, Lawrence H. Summers, and Robert J. Waldmann. "The Economic Consequences of Noise Traders." NBER Working Paper no. 2395, October 1987.
- De Long, J. Bradford, Andrei Shleifer, Lawrence H. Summers, and Robert J. Waldmann. "The Size and Incidence of the Losses from Noise Trading." *Journal of Finance* 44 (1989): 681–96.
- De Long, J. Bradford, Andrei Shleifer, Lawrence H. Summers, and Robert J. Waldmann. "Positive Feedback Investment Strategies and Destabilizing Rational Speculation." *Journal of Finance* 45 (1990): 379–95.
- De Long, J. Bradford, Andrei Shleifer, Lawrence H. Summers, and Robert J. Waldmann. "Noise Trader Risk in Financial Markets." *Journal of Political Economy* 98 (1990): 703–38.
- Fama, Eugene F. "The Behavior of Stock Market Prices." *Journal of Business* 38 (1965): 34–105.
- Fama, Eugene F., and Kenneth R. French. "Dividend Yields and Expected Stock Returns." *Journal of Financial Economics* 22 (1988a): 3–25.
- Fama, Eugene F., and Kenneth R. French. "Permanent and Temporary Components of Stock Prices." *Journal of Political Economy* 96 (1988b): 246–73.
- Figlewski, Stephen. "Subjective Information and Market Efficiency in a Betting Market." *Journal of Political Economy* 87 (1979): 75–88.
- Friedman, Milton. "The Case for Flexible Exchange Rates." In *Essays in Positive Economics*. Chicago: University of Chicago Press, 1953.

- Grossman, Sanford J., and Joseph E. Stiglitz. "On the Impossibility of Informationally Efficient Markets." *American Economic Review* 70 (1980): 393–408.
- Herzfeld, Thomas J. *The Investor's Guide to Closed-End Funds: The Herzfeld Edge*. New York: McGraw-Hill, 1980.
- Jensen, Michael C. "Agency Costs of Free Cash Flow, Corporate Finance, and Takeovers." *American Economic Review Papers and Proceedings* 76 (1986): 323–29.
- Kleidon, Allan W. "Anomalies in Financial Economics: Blueprint for Change?" *Journal of Business* 59 (1986): S469–S499.
- Kyle, Albert S. "Continuous Auctions and Insider Trading." *Econometrica* 53 (1985): 1315–35.
- Lee, Charles M. C., Andrei Shleifer, and Richard H. Thaler. "Investor Sentiment and the Closed-End Fund Puzzle." *Journal of Finance* 46 (1991): 75–109.
- Malkiel, Burton G. "The Valuation of Closed-End Investment-Company Shares." *Journal of Finance* 32 (1977): 847–59.
- Mayshar, Joram. "On Divergence of Opinion and Imperfections in Capital Markets." *American Economic Review* 73 (1983): 114–28.
- Mehra, Rajnish, and Edward C. Prescott. "The Equity Premium: A Puzzle." *Journal of Monetary Economics* 15 (1985): 145–61.
- Poterba, James M., and Lawrence H. Summers. "Mean Reversion in Stock Prices: Evidence and Implications." *Journal of Financial Economics* 22 (1988): 27–59.
- Roll, Richard. "Orange Juice and Weather." *American Economic Review* 74 (1984): 861–80.
- Samuelson, Paul A. "An Exact Consumption-Loan Model of Interest with or without the Social Contrivance of Money." *Journal of Political Economy* 66 (1958): 467–82.
- Shiller, Robert J. "Do Stock Prices Move Too Much to Be Justified by Subsequent Changes in Dividends?" *American Economic Review* 71 (1981): 421–36.
- Shiller, Robert J. "Stock Prices and Social Dynamics." *Brookings Papers on Economic Activity*, no. 2 (1984): 457–98.
- Shleifer, Andrei, and Robert W. Vishny. "Equilibrium Short Horizons of Investors and Firms." *American Economic Review Papers and Proceedings* 80 (1990): 148–53.
- Shleifer, Andrei, and Robert W. Vishny. "The Limits of Arbitrage." *Journal of Finance* 52 (1997): 35–55.
- Stein, Jeremy C. "Informational Externalities and Welfare-Reducing Speculation." *Journal of Political Economy* 95 (1987): 1123–45.
- Townsend, Robert M. "Forecasting the Forecasts of Others." *Journal of Political Economy* 91 (1983): 546–88.